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Track: Cloud Platform Development
Course: Jenkins
Lab 2

1- configure Jenkins image to run docker commands on your host docker daemon

```
Dockerfile U X
Dockerfile
1 FROM jenkins/jenkins:lts
2
3 USER root
4
5 # Install dependencies
6 RUN apt-get update -qq && \
7     apt-get install -y \
8         ca-certificates \
9         curl \
10        gnupg
11
12 # Add Docker official GPG key
13 RUN install -m 0755 -d /etc/apt/keyrings && \
14     curl -fsSL https://download.docker.com/linux/debian/gpg \
15     -o /etc/apt/keyrings/docker.asc && \
16     chmod a+r /etc/apt/keyrings/docker.asc
17
18 # Add Docker repository
19 RUN echo \
20     "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/debian bookworm stable" \
21     > /etc/apt/sources.list.d/docker.list
22
23 # Install Docker CLI
24 RUN apt-get update -qq && \
25     apt-get install -y docker-ce-cli
26
27 # Add jenkins user to docker group
28 RUN groupadd docker && usermod -aG docker jenkins
29
30 USER jenkins
```

```
bassant@fedora:~/Documents/Jenkins/Jenkins_ITI$ docker run -d \
--name jenkins \
-p 8080:8080 \
-p 50000:50000 \
-v jenkins_home:/var/jenkins_home \
-v /var/run/docker.sock:/var/run/docker.sock \
-e JAVA_TOOL_OPTIONS="-Djava.net.preferIPv4Stack=true" \
jenkins-with-docker:latest
f073f86a635bd77e7fbcd86ad8668b0535f1a758875c415a592868a4ede171fe
```

2- create CI/CD for this repo

https://github.com/mahmoud254/jenkins_nodejs_example.git

Q

Search by repository name

All content

Create a repository

Name	Last Pushed	Contains	Visibility	Scout
basabesso/nodejs-demo	2 minutes ago	IMAGE	Public	Inactive
basabesso/ubuntu-sleeper	3 months ago	IMAGE	Public	Inactive
basabesso/mysql	3 months ago	IMAGE	Public	Inactive
basabesso/custom_nginx	3 months ago	IMAGE	Public	Inactive

Repositories / nodejs-demo / General

basabesso/nodejs-demo

Last pushed 5 minutes ago · Repository size: 337.2 MB · ☆0 · ↓0

Add a description

Add a category

Docker commands

To push a new tag to this repository:

docker push basabesso/nodejs-demo:tagname

Public view

General

Tags

Image Management

Collaborators

Webhooks

Settings

Tags

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	5 minutes

See all

DOCKER SCOUT INACTIVE

Activate

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.



Jenkins / Lab2-Nodejs-Pipeline



Status



Changes



Build Now



Configure



Delete Pipeline



Stages



Rename



Pipeline Syntax



Lab2-Nodejs-Pipeline

Permalinks

- Last build (#4), 1 min 12 sec ago
- Last stable build (#4), 1 min 12 sec ago
- Last successful build (#4), 1 min 12 sec ago
- Last completed build (#4), 1 min 12 sec ago

Builds >



Filter



Today



#4 7:37 PM



#3 7:04 PM



Definition

Pipeline script



Script ?

```
1~ pipeline {
2~   agent any
3~
4~   stages {
5~
6~     stage('Preparation') {
7~       steps {
8~
9~         git 'https://github.com/mahmoud254/jenkins_nodejs_example.git'
10~
11~       }
12~     }
13~
14~     stage('Build Docker Image') {
15~       steps {
```

Definition

Pipeline script

Script ?

```

12     }
13
14     stage('Build Docker Image') {
15         steps {
16             withCredentials([usernamePassword(credentialsId: 'DockerHub', usernameVariable: 'USERNAME', passwordVariable: 'PASSWORD')]) {
17
18                 sh '''docker build -t basabesso/nodejs-demo:latest .
19                     docker login -u ${USERNAME} -p ${PASSWORD}
20                     docker push basabesso/nodejs-demo:latest
21                 '''
22             }
23         }
24     }
25
26     stage('Deployment') {

```

Pipeline script

Script ?

```

23     }
24 }
25
26     stage('Deployment') {
27         steps {
28             withCredentials([usernamePassword(credentialsId: 'DockerHub', usernameVariable: 'USERNAME', passwordVariable: 'PASSWORD')]) {
29
30                 sh '''docker login -u ${USERNAME} -p ${PASSWORD}
31                     docker run -d -p 3000:3000 basabesso/nodejs-demo:latest
32                 '''
33             }
34         }
35     }
36 }
37

```

☒ Use Groovy Sandbox ?

```

2175d13117406: Pushed
07dff4ad21eb: Pushed
latest: digest: sha256:038ed512825bd334964cf2cfbdc6fb6b01e82717fc5107a48438367492a459b2 size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

```

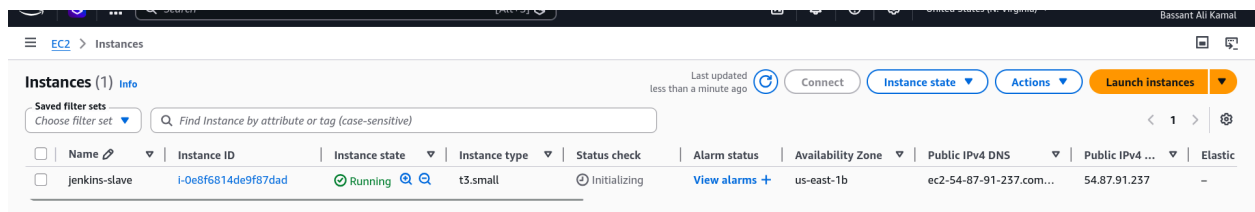
bassant@fedora:~/Documents/Jenkins/Jenkins_ITI$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
9fd45408a6c0   basabesso/nodejs-demo:latest        "docker-entrypoint.s..." 2 minutes ago  Up 2 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
p073f86a635b   jenkins-with-docker:latest          "/usr/bin/tini -- /u..." 38 minutes ago  Up 38 minutes  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
p, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   jenkins
bassant@fedora:~/Documents/Jenkins/Jenkins_ITI$

```



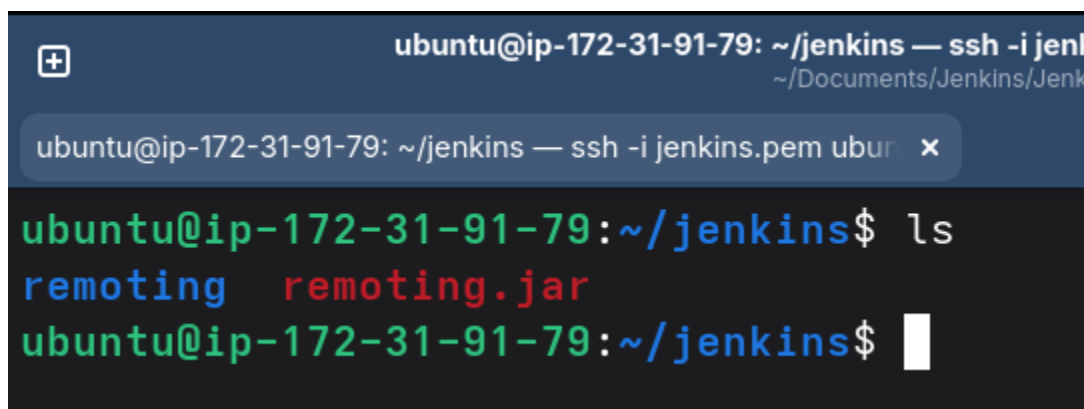
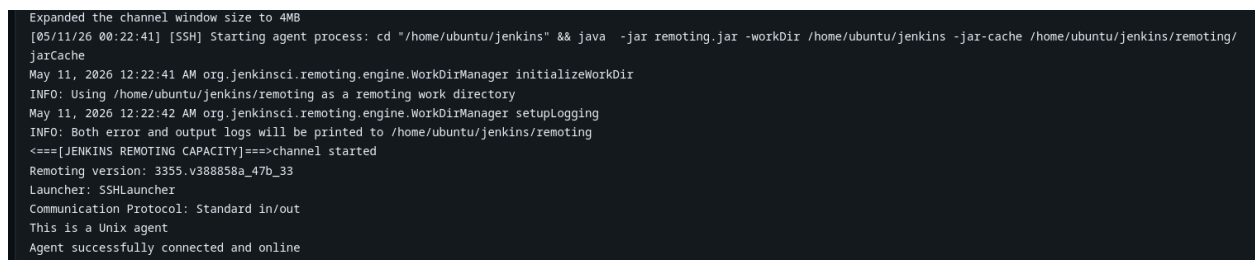
Hello World from ITI 46

3- create ec2 in aws and configure it as Jenkins slave



Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	aws-slave	Linux (amd64)	3.8 sec ahead	15.61 GiB	0 B	15.61 GiB	8400ms
	Built-In Node	Linux (amd64)	In sync	133.15 GiB	8.00 GiB	133.15 GiB	0ms
	Data obtained	1 min 11 sec	1 min 9 sec	1 min 2 sec	1 min 9 sec	1 min 2 sec	1 min 8 sec



```

26
27~   stage('Deployment') {
28     agent { label 'aws' }
29~   steps {
30~     withCredentials([usernamePassword(credentialsId: 'DockerHub', usernameVariable: 'USERNAME', passwordVariable: 'PASSWORD')])
31
32     sh '''
33         docker login -u ${USERNAME} -p ${PASSWORD}
34         docker run -d -p 3000:3000 basabesso/nodejs-demo:latest
35     '''
36   }
37 }
38 }
39 }
40

```

```

[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deployment)
[Pipeline] node
Running on aws-slave in /home/ubuntu/jenkins/workspace/Lab2-Nodejs-Pipeline
[Pipeline] {
[Pipeline] withCredentials

```

```

[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: bassantalikamal, channel: iti46, color: good, botUser: false, tokenCredentialId: Slack,
notifyCommitters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

```

ubuntu@ip-172-31-91-79: ~/jenkins -- ssh -i jenkins.pem ubuntu@54.87.91.237
ubuntu@ip-172-31-91-79:~/jenkins$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
ab92054a1413   basabesso/nodejs-demo:latest        "docker-entrypoint.s..." 2 minutes ago  Up 2 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gifted_poitras
ubuntu@ip-172-31-91-79:~/jenkins$

```

sg-01cf6eac6cbe4489f - launch-wizard-1

Inbound security group rules successfully modified on security group (sg-01cf6eac6cbe4489f | launch-wizard-1)

Details

sg-01cf6eac6cbe4489f - launch-wizard-1

Details

Security group name launch-wizard-1	Security group ID sg-01cf6eac6cbe4489f	Description launch-wizard-1 created 2026-05-10T23:54:28.919Z	VPC ID vpc-0c05508e28087ef89
Owner 707938860152	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

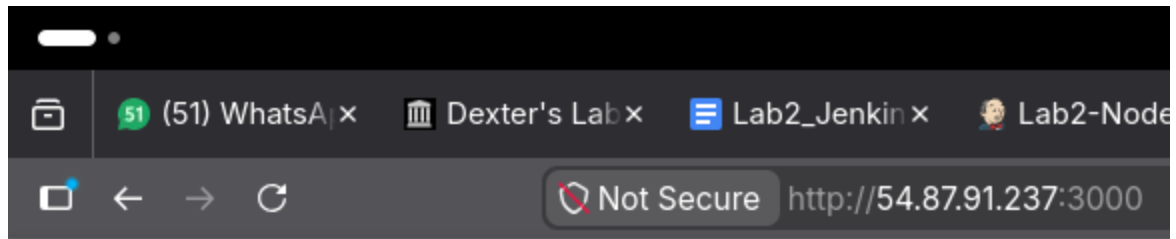
[Inbound rules](#) |
 [Outbound rules](#) |
 [Sharing](#) |
 [VPC associations](#) |
 [Related resources - new](#) |
 [Tags](#)

Inbound rules (2)

[Manage tags](#) [Edit inbound rules](#)

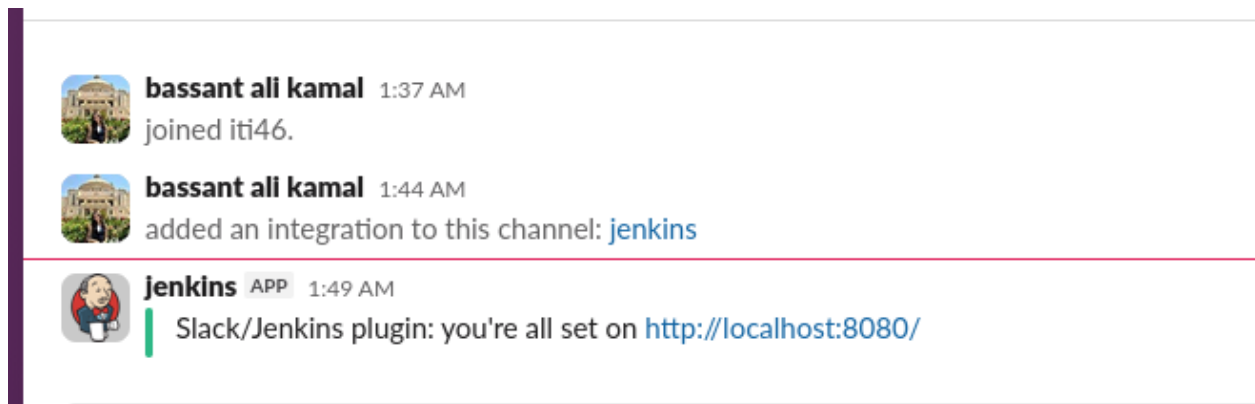
Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-012d147fda62a20c8	IPv4	Custom TCP	TCP	3000	0.0.0.0/0
☐	-	sgr-0c22d8a33110b5806	IPv4	SSH	TCP	22	0.0.0.0/0

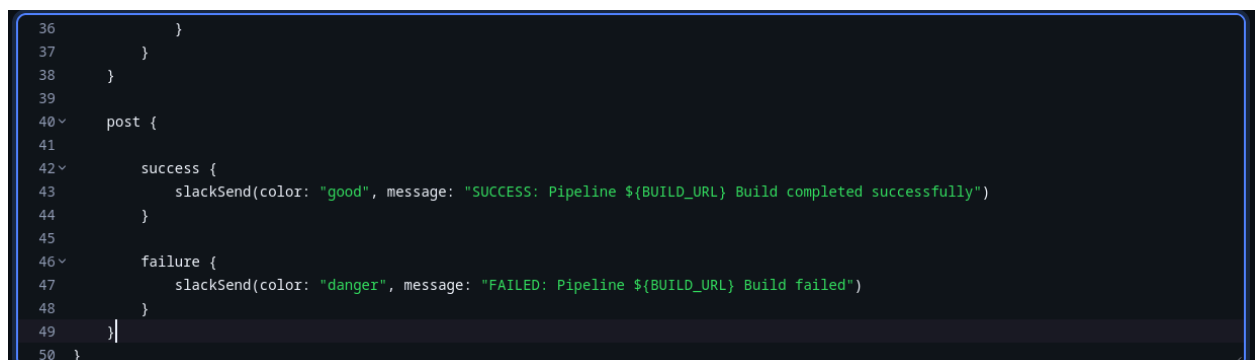


Hello World from ITI 46

4- integrate slack with jenkins



5- send slack message when stage in your pipeline is successful or failed



Slack

Workspace ?

bassantalikamal

Credential ?

Slack Token

Default channel / member id ?

iti46

☐ Custom slack app bot user ?

Advanced ▼

```
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint determined_bardeen (48b906a94c2674929675db31d301063219e5f1335e86def711f957c901434028): Bind for 0.0.0.0:3000 failed: port is already allocated
```

```
Run 'docker run --help' for more information
```

```
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: bassantalikamal, channel: iti46, color: danger, botUser: false, tokenCredentialId: Slack,
notifyCommitters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
ERROR: script returned exit code 125
Finished: FAILURE
```

```
Login Succeeded
```

```
+ docker run -d -p 3000:3000 basabesso/nodejs-demo:latest
608e899b49e8899f557cdcc7b2c858b30d6362e84991343a2531be666b4acdda
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: bassantalikamal, channel: iti46, color: good, botUser: false, tokenCredentialId: Slack,
notifyCommitters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



basant ali kamal 1:37 AM

joined iti46.



basant ali kamal 1:44 AM

added an integration to this channel: [jenkins](#)



jenkins APP 1:49 AM

Slack/Jenkins plugin: you're all set on <http://localhost:8080/>



jenkins APP 2:44 AM

FAILED: Pipeline <http://localhost:8080/job/Lab2-Nodejs-Pipeline/7/> Build failed

2:46

SUCCESS: Pipeline <http://localhost:8080/job/Lab2-Nodejs-Pipeline/8/> Build completed successfully

6- install audit logs plugin and test it

The screenshot shows the Jenkins 'Plugins' page. The search bar contains 'aud'. The 'Audit Log Plugin 1.3' is listed with a green 'ON' status and a toggle switch set to 'ON'. The description reads: 'Audit logging for Jenkins with predefined audit events. Report an issue with this plugin.'

Log session start time 2026-05-11T01:28:19.411Z

Time	Thread	Level	Logger	Message
2026-05-11T01:28:19.47Z	Handling POST /j_spring_security_check from 172.17.0.1 : Jetty (winstone)-61	OFF	AuditLogger	Audit [login timestamp="2026-05-11T01:28:19.467Z" userId="admin"]
2026-05-11T01:32:16.426Z	Handling POST /view/all/createItem from 172.17.0.1 : Jetty (winstone)-30	OFF	AuditLogger	Audit [createItem itemName="test-pipeline" itemUri="/job/test-pipeline/" timestamp="2026-05-11T01:32:16.424Z" userId="admin"]
MDC: {requestUri=/view/all/createItem, userId=admin, requestId=3154cf3-a764-4673-86b2-500ed0575009, requestMethod=POST}				
2026-05-11T01:32:20.458Z	Handling POST /job/test-pipeline/configSubmit from 172.17.0.1 : Jetty (winstone)-29	OFF	AuditLogger	Audit [updateItem itemName="test-pipeline" itemUri="/job/test-pipeline/" timestamp="2026-05-11T01:32:20.458Z" userId="admin"]
MDC: {requestUri=/job/test-pipeline/configSubmit, userId=admin, requestId=1248c390-b1da-4652-94c0-03b063c6ae9a, requestMethod=POST}				
2026-05-11T01:33:39.696Z	Handling POST /job/test-pipeline/delete from 172.17.0.1 : Jetty (winstone)-28	OFF	AuditLogger	Audit [deleteItem itemName="test-pipeline" itemUri="/job/test-pipeline/" timestamp="2026-05-11T01:33:39.695Z" userId="admin"]
MDC: {requestUri=/job/test-pipeline/delete, userId=admin, requestId=efac5c5-1036-44e3-9cb5-f78fbf95a24, requestMethod=POST}				

7- fork the following repo

https://github.com/mahmoud254/Booster_CI_CD_Project
and add dockerfile to run this django app and use github actions to build the docker image and push it to your Dockerhub



Code Blame 16 lines (9 loc) · 262 Bytes



```
1 FROM python:3.6
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6
7 RUN pip install --no-cache-dir -r requirements.txt
8
9 COPY . .
10
11 RUN python3.6 manage.py makemigrations
12 RUN python3.6 manage.py migrate
13
14 EXPOSE 8000
15
16 CMD ["python3.6", "manage.py", "runserver", "0.0.0.0:8000"]
```



Besso2003 Update Docker workflow to include login and push steps ✓

Code

Blame

25 lines (18 loc) · 506 Bytes

```
1  name: Docker Image CI
2
3  on:
4    push:
5      branches: [ "master" ]
6
7  jobs:
8    build:
9
10     runs-on: ubuntu-latest
11
12     steps:
13       - uses: actions/checkout@v4
14
15       - name: Login to DockerHub
16         uses: docker/login-action@v3
17         with:
18           username: ${ secrets.DOCKER_USERNAME }
19           password: ${ secrets.DOCKER_PASSWORD }
20
21       - name: Build Docker Image
22         run: docker build -t basabesso/booster-django:latest .
23
24       - name: Push Docker Image
25         run: docker push basabesso/booster-django:latest
```

Repository secrets

New repository secret

Name 	Last updated
 DOCKER_PASSWORD	7 minutes ago  
 DOCKER_USERNAME	9 minutes ago  

← Docker Image CI

✓ Update Docker workflow to include login and push steps #1 Re-run all jobs Latest #2 ...

Summary

All jobs

✓ build

Run details

Usage

Workflow file

build
succeeded now in 3m 28s

Search logs

- > ✓ Set up job 2s
- > ✓ Run actions/checkout@v4 1s
- > ✓ Login to DockerHub 1s
- > ✓ Build Docker Image 3m 13s
- > ✓ Push Docker Image 8s
- > ✓ Post Login to DockerHub 1s
- > ✓ Post Run actions/checkout@v4 0s
- > ✓ Complete job 0s

Name	Last Pushed ↑	Contains	Visibility	Scout
basabesso/ booster-django	less than a minute ago	IMAGE	Public	Inactive
basabesso/ nodejs-demo	1 day ago	IMAGE	Public	Inactive
basabesso/ ubuntu-sleeper	3 months ago	IMAGE	Public	Inactive
basabesso/ mysql	3 months ago	IMAGE	Public	Inactive
basabesso/ custom_nginx	3 months ago	IMAGE	Public	Inactive

1-5 of 5 < >

Besso2003 / **Booster_CI_CD_Project** Public Pin Watch 0

forked from [mahmoud254/Booster_CI_CD_Project](#)

master 2 Branches 0 Tags Go to file Add file Code

This branch is up to date with [mahmoud254/Booster_CI_CD_Project:master](#) Contribute Sync fork

mahmoud254 Update README.md 54ef3c3 · 5 years ago 11 Commits

simpleApp	first commit	6 years ago
README.md	Update README.md	5 years ago
manage.py	first commit	6 years ago
requirements.txt	Update requirements.txt	6 years ago

8- Create infrastructure pipeline to run terraform with Jenkins Task

Jenkinsfile

```
1 pipeline {
2     agent any
3
4     stages {
5
6         stage('Clone Repository') {
7             steps {
8                 git 'https://github.com/Besso2003/Terraform_ITI.git'
9             }
10        }
11
12        stage('Terraform Init') {
13            steps {
14                dir('Labs') {
15                    withCredentials([
16                        [$class: 'AmazonWebServicesCredentialsBinding',
17                         credentialsId: 'aws-creds']
18                    ]) {
19                        sh 'terraform init'
20                    }
21                }
22            }
23        }
24    }
```

```
stage('Terraform Plan') {
  steps {
    dir('Labs') {
      withCredentials([
        [$class: 'AmazonWebServicesCredentialsBinding',
         credentialsId: 'aws-creds'],
        file(credentialsId: 'prod-tfvars', variable: 'TFVARS')
      ]) {
        sh 'terraform plan -var-file=$TFVARS'
      }
    }
  }
}

stage('Terraform Apply') {
  steps {
    input message: "Approve Apply?"
    dir('Labs') {
      withCredentials([
        [$class: 'AmazonWebServicesCredentialsBinding',
         credentialsId: 'aws-creds'],
        file(credentialsId: 'prod-tfvars', variable: 'TFVARS')
      ]) {
        sh 'terraform apply -auto-approve -var-file=$TFVARS'
      }
    }
  }
}
```

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/Besso2003/Terraform_ITI.git

Credentials ?

- none -

Advanced ▾

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/master

+ Add

Script Path ?

Labs/Jenkinsfile



Lightweight checkout ?

```
[0m[1m[32m
Apply complete! Resources: 2 added, 0 changed, 1 destroyed.
[0m[0m[1m[32m
Outputs:
```

```
[0menvironment = "prod"
region = "eu-north-1"
vpc_cidr = "vpc-0ef7bc7a0424ff175"
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // dir
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

4 minutes ago

Saved filter sets
Choose filter set ▾

Find Instance by attribute or tag (case-sensitive)

< 1

☐	Name 🔗 ▾	Instance ID	Instance state ▾	Instance type ▾	Status check	Alarm status	Availability Zone ▾	Public IPv4
☐	prod-bastion	i-06595b1c5417cd72c	Running 🔍 🔍	t3.small	3/3 checks passed	View alarms +	eu-north-1a	-
☐	prod-app	i-0058c69b2f410a9ba	Running 🔍 🔍	t3.small	3/3 checks passed	View alarms +	eu-north-1a	-

All +

S	W	Name ↓	Last Success	Last Failure	Last Duration	
✓	☁	Lab2-Nodejs-Pipeline	1 day 20 hr #11	1 day 20 hr #9	31 sec	▶
✓	☁	Terraform-Pipeline	4 min 35 sec #18	14 min #17	42 sec	▶